

Has Idiomaticity Representation Improved? A Controlled, Model-Agnostic Re-Evaluation Across Model Generations

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Abstract. He et al. (2025) showed that 2024-era word- and sentence-representation models do not faithfully encode idiomaticity, relying instead on lexical overlap. We ask whether the rapid progress of language models since then has changed this, and if not, what the underlying obstacle is. We re-implement the original controlled evaluation—Noun Compound Idiomaticity Minimal Pairs (NCIMP), the four probes, and the Affinity / Scaled-Similarity metrics—as a model-agnostic framework, validate it against the published results, and apply it across three cohorts: paper-era multilingual encoders, modern encoder and sentence-embedding models, and an eleven-model Qwen/Gemma decoder cohort from 2025–2026. Across twenty model conditions, no model reaches the partial-capture threshold. The best model, base Gemma 4-12B, reaches $ICS = 0.525$ and is only the first to prefer the gold idiomatic synonym over the literal component. We also introduce the Ordinary-Calibrated Gap (OCG), which normalises idiomatic synonym-vs-random separation by a model’s behaviour on ordinary single-word and two-word controls. The calibration shows that models separate ordinary synonyms from random replacements much more cleanly than they separate idioms from literal alternatives. The results support a structural-bottleneck account: scale narrows, but does not solve, idiomaticity representation.

Key-words: Idiomaticity; multiword expressions; sentence embeddings; model evaluation; anisotropy; UniDive.

Table 1. English, paper-era vs modern encoder cohort (NC level).

Indicator	Paper-era	Modern	Δ
ICS (composite $\uparrow$)	0.341	0.430	+0.089 (+26%)
ISC ($\uparrow$)	0.099	0.174	+0.075
IG ($\downarrow$)	0.446	0.348	-0.098
LOD ($\downarrow$, < 0)	0.227	0.077	-0.150
AID ($\uparrow$, > 0)	-0.108	-0.021	+0.087
FLOOR (anisotropy)	0.575	0.723	+0.148
RHO ($\downarrow$)	0.615	0.578	-0.037

Table 2. Portuguese, paper-era vs modern encoder cohort (NC level).

Indicator	Paper-era	Modern	Δ
ICS ($\uparrow$)	0.428	0.423	-0.005 ($\approx 0\%$)
ISC ($\uparrow$)	0.150	0.095	-0.055
IG ($\downarrow$)	0.304	0.328	+0.024
LOD ($\downarrow$)	0.203	0.158	-0.045
AID ($\uparrow$)	-0.085	-0.026	+0.059
FLOOR	0.608	0.730	+0.122
RHO ($\downarrow$)	0.421	0.492	+0.072

1. Introduction

2. Background and Related Work

3. Method

4. The Ordinary-Calibrated Gap (OCG)

5. Results

6. Discussion

7. Conclusion and Future Work

Use of AI and AI-assisted technologies